

As Soviet propagandists crowed over the Soviet Union's technological achievement, claiming it as evidence of the superiority of the communist system, the United States raced to catch up in what had instantly become a "space race." Both the USSR and the US had announced a satellite launch for International Geophysical Year 1957–58, developing a launcher named Vanguard. It was in no way comparable to the Soviet R-7 in thrust or payload, but its first

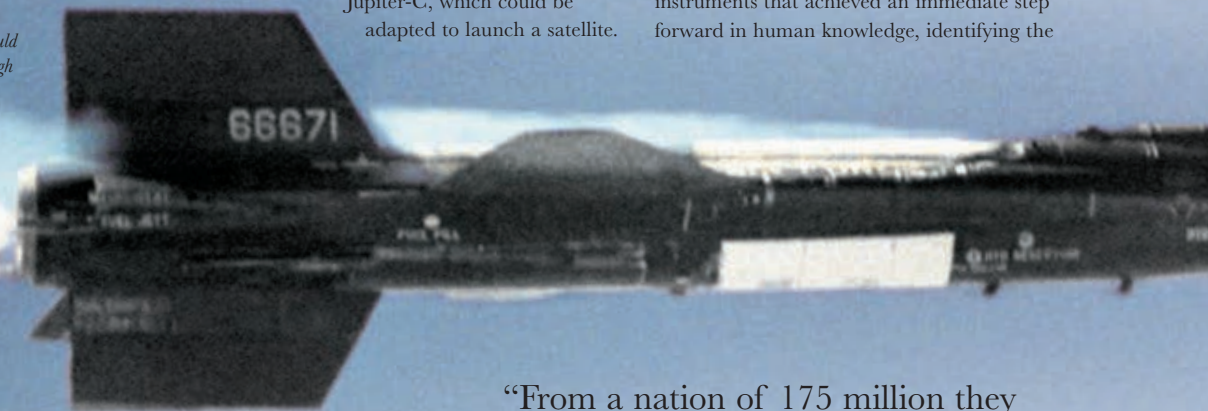
launch in December 1957 was eagerly awaited as an urgently needed riposte to the Sputniks. But Vanguard blew up on liftoff – the explosion seeming to one observer "as if the gates of hell had opened up." The delighted Soviets sneeringly inquired whether America might like to benefit from technical aid for backward countries.

Fortunately, Vanguard was not the only satellite project the United States had in hand. Wernher von Braun's team in Huntsville, working for the US Army, had developed a more powerful version of their Redstone rocket, the Jupiter-C, which could be adapted to launch a satellite.

Although interservice rivalries had caused the Army's long-range rocket program to be curtailed, when the shock of the Sputnik launches struck, von Braun and his boss, General John Medaris, were ready to step forward with their own proposal for a satellite launch. The Jupiter-C rocket was adapted into a Juno 1 launcher, while the Jet Propulsion Laboratory in California, headed by William Pickering, put together a satellite to go on top of it. Explorer 1 was launched into orbit on January 31, 1958. Unlike the first Soviet satellites, it carried scientific instruments that achieved an immediate step forward in human knowledge, identifying the

SPACE AIRCRAFT

The North American X-15 rocket plane could fly to an altitude of 67 miles (108km) – high enough to qualify as suborbital space flight. But it was not capable of going into orbit, and even if it had been, it would not have been able to reenter the atmosphere safely.



"From a nation of 175 million they stepped forward... seven men cut of the same stone as Columbus..."

TIME MAGAZINE

AFTER THE FIRST MERCURY 7 PRESS CONFERENCE

AMERICA'S FIRST ASTRONAUTS

The Mercury 7 pose for a formal portrait in their space gear in 1962: (left to right) Wally Schirra, Alan Shepard, Deke Slayton, Gus Grissom, John Glenn, Gordon Cooper, and Scott Carpenter. Selection for the space program had made these pilots among the most famous men in the world.



regions of radiation trapped by the Earth's magnetic field that are now known as the Van Allen Belts – after physicist James Van Allen, who set up the experiment conducted by Explorer 1.

But the pursuit of scientific knowledge or of practical technological goals such as improved communications were to remain for a long time marginal to the main thrust of both American and Soviet space programs. The space race was a contest for prestige between two superpowers, each seeking to prove the superiority of its own political and economic system on a cosmic stage. Both sides knew that in the eyes of the world, the winner would be the first to put a man in space. The imperative of the moment was not long-term planning or practical utility, but simply to put a man in space as soon as possible.

During 1958, the United States sorted out the basic organization of its space effort into three programs. Determined that one program should be